AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

MODEL EXAMINATION PAPER 2026 AND ONWARDS

Physics Paper II

Time: 1 hour 30 minutes Marks: 35

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INSTRUCTIONS:

1. Check your name and school information above. Sign if the information is correct.

I agree that this is my name and school. **Candidate's Signature**

2. Read each question carefully.

3. There are TWELVE questions in this exam. Answer ALL questions. Questions 11 & 12 each offer TWO choices. For each of these questions, attempt ONE choice only.

4. When answering the questions:
   - Use a black pen/ pointer to write your answers. DO NOT write your answers in pencil.
   - Use a black pencil for diagrams and drawings. DO NOT use coloured pencils, markers or highlighters.
   - DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
     If you make a mistake, cross it out neatly and write the correct answer beside it.
   - Write your answers only in the allocated space provided for each question.
   - DO NOT write outside the answer space. Answers written outside the allocated space will not be marked.

5. The marks for each question are shown in brackets ( ) next to the question number. Use this information to manage your time efficiently.

6. DO NOT mention your name or school name anywhere inside the paper.

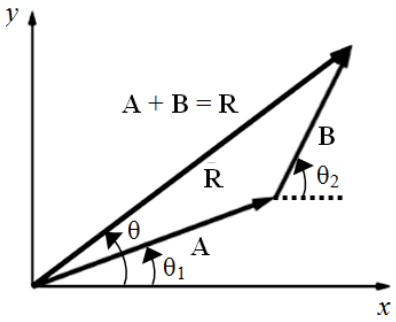
7. You may use a scientific calculator if you wish

Q.1. (Total 2 Marks)

Show that $2aS = v_f^2 - v_i^2$ is dimensionally correct.

Q.2. (Total 2 Marks)

The given diagram shows the resultant vector **R** of two vectors **A** and **B**.



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Write the formula for each of the following.

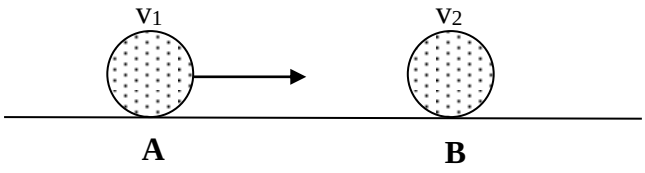
i. Magnitude of the vector **A** (1 Mark)

ii. Direction of the resultant vector **R** (1 Mark)

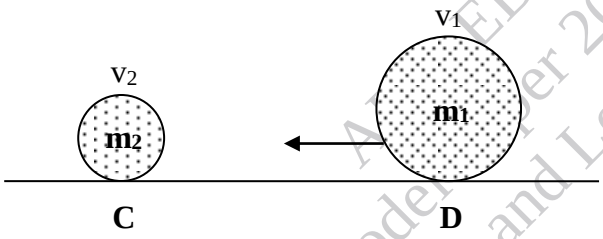
Q.3. (Total 2 Marks)

Consider two smooth, non-rotating balls colliding elastically in an isolated system. What will be the effect on the velocity and directions of the balls if

i. ball **A** moving in a straight line collides with the identical, stationary ball **B** of equal mass? (1 Mark)



ii. heavy ball **D** moving in a straight line collides with a lighter ball **C** at rest? (1 Mark)



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Q.4. (Total 3 Marks)

Derive the relation for the total time period of a projectile.

Q.5. (Total 2 Marks)

A student is stretching a rubber band to tie his colour pencils.

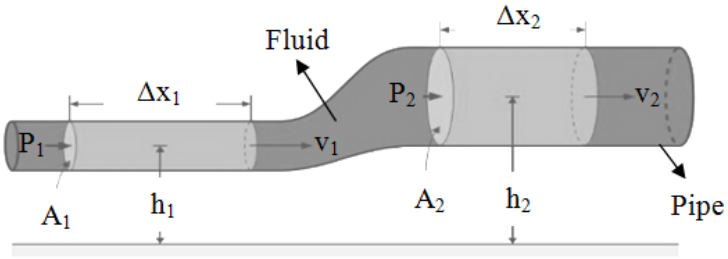
i. Identify the quantities that define the applied force and the resulting change in length. (1 Mark)

ii. Compare the strain in a rubber band to that in a metal wire under similar stress. (1 Mark)

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Q.6. (Total 2 Marks)

i. By looking at the given diagram of a pipe, write an appropriate equation for the fluid flow. (1 Mark)



ii. To understand the features of fluids in motion, the behaviour of fluids should satisfy three conditions. State any ONE of these conditions. (1 Mark)

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Q.7. (Total 3 Marks)

Calculate the frequency of a simple pendulum, if its length is 50 cm.

(**Note**: The acceleration due to gravity 'g' is 9.8 m/s^2.)

Q.8. (Total 3 Marks)

Two ambulances, 'A' and 'B' are moving with the velocity of 20 m/s with the normal siren frequency of 650 Hz. Ambulance 'A' moves towards a stationary listener while ambulance 'B' moves away from the stationary listener.

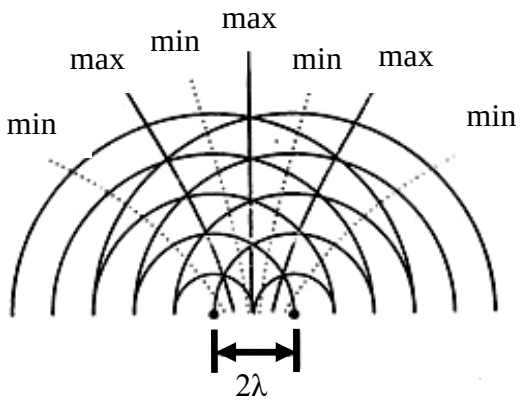
Calculate the frequency of the siren heard by the stationary listener from each ambulance.

(**Note**: The speed of sound in the air is 340 m/s.).

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Q.9. (Total 3 Marks)

i. Identify the phenomenon of light shown in the given diagram. (1 Mark)



ii. Identify any TWO conditions required for the occurrence of this phenomenon. (2 Marks)

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Q.10. (Total 3 Marks)

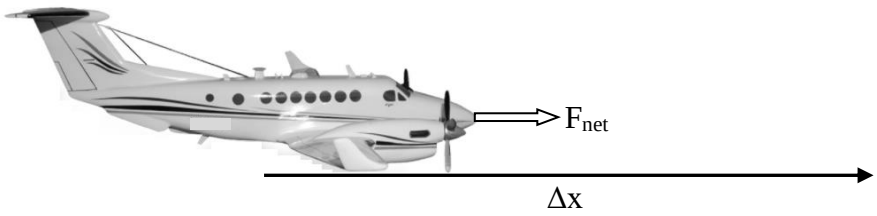
Oceanologists are discussing the difference in water temperature between the surface and the bottom of the ocean for operating a heat engine.

If they observed 20°C as high temperature and 4°C as low temperature, then calculate the efficiency in percentage of the heat engine.

Q.11. (Total 5 Marks)

EITHER

a. A constant force (F_{net}) acts on an airplane of mass (m) and it covers a distance (Δx) with an increase in its velocity from (v_i) to (v_f).



Deduce an equation of the work-energy principle for the given situation. (5 Marks)

OR

b. A missile is fired with a velocity of 450 m/s at an angle of 45° with the horizontal axis.

Calculate

i. the range of the projectile. (2 Marks)

ii. the time for which the missile will remain in the air. (3 Marks)

(**Note**: The acceleration due to gravity as 10 m/s^2.)

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Q.12. (Total 5 Marks)

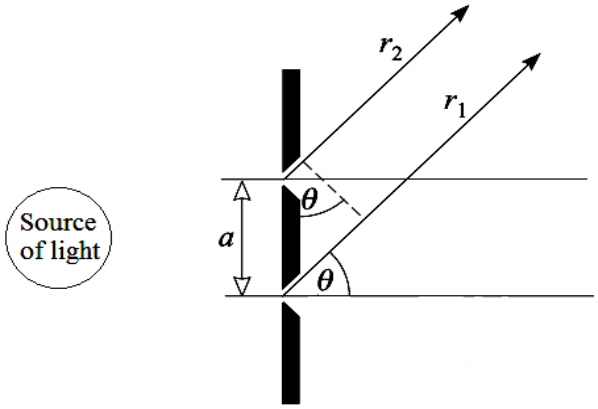
EITHER

a. A 2.4 m string vibrates in its 4th harmonic with a wave speed of 96 m/s.

Calculate wavelength, frequency, nodes, antinodes, and draw the wave pattern.

OR

b. Two narrow horizontal, parallel slits at a distance *a* are illuminated by a light beam of 500 nm as shown in the given figure.



i. Complete the following using symbols from the given diagram. (2 Marks)

I. Difference in path lengths for the two beams (in term of *r*) =_____

II. $\sin\theta$ = _____

ii. If a = 0.60 mm, find two smallest values of θ where reinforcement of light waves occurs. (3 Marks)

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